



Automated diabetic retinopathy detection with two different retinal imaging devices using artificial intelligence: a comparison study

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Abstract

Purpose In this study, we evaluated the diagnostic performance of an automated artificial intelligence-based diabetic retinopathy (DR) algorithm with two retinal imaging systems using two different technologies: a conventional flash fundus camera and a white LED confocal scanner.

Methods On the same day, patients underwent dilated colour fundus photography using both a conventional flash fundus camera (TRC-NW8, Topcon Corporation, Tokyo, Japan) and a fully automated white LED confocal scanner (Eidon, Centervue, Padova, Italy). All images were analysed for DR severity both by retina specialists and the AI software EyeArt (Eyenuk Inc., Los Angeles, CA) and graded as referable DR (RDR) or not RDR. Sensitivity, specificity and the area under the curve (AUC) were computed.

Results A series of 165 diabetic subjects (330 eyes) were enrolled. The automated algorithm achieved 90.8% sensitivity with 75.3% specificity on images acquired with the conventional fundus camera and 94.1% sensitivity with 86.8% specificity on images obtained from the white LED confocal scanner. The difference between AUC was 0.0737 ($p = 0.0023$).

Conclusion The automated image analysis software is well suited to work with different imaging technologies. It achieved a better diagnostic performance when the white LED confocal scanner is used. Further evaluation in the context of screening campaigns is needed.

Keywords Artificial intelligence · Conventional flash fundus camera · Deep learning · Diabetic retinopathy · Telemedicine · White LED confocal scanner

Introduction

Diabetic retinopathy (DR) is a primary complication of diabetes mellitus, and it is one of the main causes of blindness among working-age adults. With over 400 million people with diabetes worldwide, the overall prevalence of DR is 34.6%, with 10.2% of them having sight-threatening DR [1]. Early diagnosis of DR requires systematic and extensive screening due to the frequently asymptomatic nature of this major cause of preventable blindness [2]. Fundus photography has been accepted as the current mode of screening for DR with grading

of the fundus photographs being done by trained certified graders or retinal specialists [3]. With the exponential increase in the number of people with diabetes and shortage of trained graders/retinal specialists, an automated approach using artificial intelligence (AI) for DR screening has started attracting attention of healthcare professionals. AI is a machine-based tool that may improve healthcare systems' efficiency. A subset of AI is deep learning (DL) that is a specific type of artificial neural network that resembles the multilayered human cognition system. Algorithms based on a DL approach are applied in the analysis of complex medical data to provide useful diagnostic outputs. AI and DL find applications in ophthalmology as a large part of data to be analysed is image-based and the outcomes are related to image recognition and subsequent interpretation. Multiple efforts have been made in order to develop DL algorithms for the automated detection of DR lesions from colour fundus photographs. When implemented in screening programs, these systems have the potential to reduce costs and increase service efficiency in developed world healthcare economies and to aid delivering DR

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screening programs in developing and remoting settings [4, 5].

Several studies have reported the diagnostic performance of AI-based software in the detection of referable DR (RDR), which is usually defined as a grade of retinopathy equal or more severe than moderate non-proliferative DR, with and without diabetic macular oedema (DME). The published results appear promising because they show a sensitivity ranging from 75 to 94.7%, with a specificity between 50 and 95% [6–12].

Similar to human performance, automated image analysis software may be influenced by image resolution, contrast, and fidelity to the retinal appearance. With the aim to train itself, a DL neural network is dependent upon having a variable and large enough dataset available. Different public datasets (Messidor, EyePACS, Kaggle) are currently used for the development and validation of DL algorithms. These datasets are composed of millions of colour retinal images, acquired in multi-ethnic populations and with different models of conventional flash fundus camera [13–16].

In a previous study, we showed that a white light-emitting diode (LED) confocal scanner (Eidon, Centervue, Padova, Italy) provides retinal images that significantly differ from those acquired by a conventional flash fundus camera, having a greater discriminative power, diagnostic accuracy and providing an accurate documentation of retinal appearance [17]. In this study, we tested an automated DL-based DR image analysis software with two different retinal imaging devices, the above-mentioned white LED confocal scanner and a conventional flash fundus camera.

Methods

Study population

Patients with type 1 or type 2 diabetes mellitus who attended a routine annual visit at the ambulatory surgical centre Istituto Europeo di Microchirurgia Oculare, IEMO, were invited to participate in the study. The study protocol follows the tenets of the Declaration of Helsinki. All participants were 18 years or older and provided written, informed consent. Demographic and clinical details of recruited patients were collected. Visual acuity was measured using Snellen charts.

Image acquisition

On the same day, each patient underwent dilated colour fundus photography using both a conventional flash fundus camera (TRC-NW8, Topcon Corporation, Tokyo, Japan) and a fully automated, white LED confocal scanner (Eidon, Centervue, Padova, Italy). The conventional system is a high-definition fundus camera that captures 45-degree 12-

megapixel digital images through pupils as small as 3.3 mm and uses a xenon light source. The scanner is a slit confocal system that acquires 60-degree 14-megapixel retinal images in an automated fashion through a non-mydratic pupil (inferior to 2.5 mm) using a broad-spectrum white LED as light source.

According to the local protocol, one fundus photographic field which includes the macula and arcades (macula-centred field) was captured for each eye. Images were obtained by a trained technician. Care was taken to generate gradable quality images. According to the UK screening programme, images are gradable when their acquisition is of adequate quality. A macular image is classified as adequate if the foveal region is situated more than two disc diameters from the edge of the image. In addition, fine vessels should be visible within one disc diameter of foveal centre. Since the two devices produce images with different angles of view, all images acquired with the white LED confocal scanner were cropped to 45° in order to have the same retinal field size for evaluation. The fundus photographs of each participant were analysed by two masked ophthalmologists and then transmitted to the AI software.

Human grading

All fundus photographs were graded independently by two retina specialists (VS and DV) who were masked to each other and to any patient records. The human graders were allowed to evaluate images from both systems (45°) simultaneously.

Graders were permitted to optimize the image quality to their satisfaction by adjusting image colour, contrast, brightness and gamma correction, as well as magnification. When the results between the two retina specialists were inconsistent, a third retina specialist (PL) would make a decision.

A diabetic retinopathy severity score was assigned according to the grading scheme proposed by the UK National Health Service (NHS) screening programme [18]. Patients without DR are graded as R0. Patients with microaneurysms, haemorrhages and/or cotton wools and/or hard exudates are classified as background retinopathy (R1). The R2 grade (pre-proliferative retinopathy) consists of the characteristics of R1 plus venous beading and/or loop or reduplication and/or definite intraretinal microvascular abnormality. Proliferative DR is graded as R3 and includes patients with new vessels on the disc, new vessels elsewhere, vitreous or preretinal haemorrhage and retinal detachment, even in the presence of laser treatment (R3). Diabetic maculopathy is classified into two wide categories, namely, M0 for those without referable features and M1 with referable features. Referable features include any hard exudates within one disc diameter of the fovea or an area of hard exudates in the macular region or presence of microaneurysm/haemorrhages within one disc diameter of the macula with a visual acuity of 20/40 ETDRS Snellen equivalent or less. RDR was defined as the presence of R2 or higher, presence of DME or both. Ungradable images (U)

were classified as technical failures because of out of focus or obscured images by glare or motion artefact. Retinal photographs with insufficient quality were also flagged as ungradable if the images had more than a third of the photograph obscured. Images deemed ungradable for human graders were not included in the AI system analysis.

Automated grading

Only images that were rated gradable by the human graders were transmitted for AI-based analysis. After masking the identity of the patient, retinal images were uploaded to Eyenuk Inc. (Los Angeles, CA, USA) for the analysis with the EyeArt software v2.1 (Eyenuk Inc., Los Angeles, CA). Cropped images were analysed exactly as they were outputted from the two fundus imaging systems. No image processing (e.g. tone and contrast enhancing/adjustment, colour normalization) was performed before providing the images for EyeArt analysis.

EyeArt is a computerized, cost-effective, cloud-based software that can automatically analyse retinal images and provide DR severity and recommendation by automatically detecting the presence, size, position and number of DR lesions. It has received the Conformité Européenne Mark and is commercially available in the European Union. EyeArt was trained and validated on colour retinal images of over 80,000 patients acquired with conventional flash fundus cameras. The version 2.1 of the software combines core DR analysis algorithms, including those for image enhancement, interest region identification, descriptor computation, with an ensemble of deep artificial neural networks for DR classification, and detection of maculopathy. The algorithm reports not only the DR severity level on the UK NHS severity scale but also the presence/absence of diabetic maculopathy. The software is designed to identify cases of DR and provides an output of ‘disease’ or ‘no disease’. An additional output of ‘refer’ versus ‘no refer’ is designed to identify cases which require referral to eye hospital services [19].

In the present study, the EyeArt algorithm was applied to automatically determine the presence of RDR, defined as the presence of R2 grade or higher and/or maculopathy. EyeArt software does not take the visual acuity as an input while it flags maculopathy as the presence of surrogate marker for DME.

Statistical analysis

Receiver operator characteristic curves (ROC) were calculated to demonstrate the association between sensitivity and specificity for the detection of RDR using the AI-based algorithm with both conventional fundus camera and the white LED confocal scanner. The performance of the AI algorithm was evaluated using the retina specialist grading as a reference

standard. The ROC curves give a measure of the accuracy of the grading system. The accuracy is represented by the area under the curve (AUC). Perfect agreement with the gold standard (human grading) would be indicated by AUC = 1. The closer the curve converges towards the top left-hand corner, the more accurate the AI-based system is. Statistical inference to compare the two ROC curves was performed using the method of DeLong et al. [20]. Statistical analyses were performed using MedCalc, version 15.0 (MedCalc Software, Ostend, Belgium) and SPSS statistical package, version 25 (IBM Corp, Armonk, NY, USA). The sample size was determined to detect a difference of 0.05 in ROC AUC. With alpha 0.05 (two-sided) and beta 0.2, the calculated sample size was at least 300 images per group, with a conservative correlation value of 0.4.

Results

A consecutive series of 165 patients with diabetes mellitus (330 eyes) were recruited for the study at the Istituto Europeo di Microchirurgia Oculare, IEMO, Udine, Italy. Patients’ characteristics are shown in Table 1.

A total of 660 retinal images were obtained and graded in this study. Images from 8 eyes (2.4%) were classified as ungradable by the human graders and thus were not considered for comparison with the automated assessment. The reasons were mainly due to inadequate field capture (5 eyes) and to the presence of significant media opacities (3 eyes).

Out of the remaining 322 eyes, no visible DR was found in 30.1% of eyes (R0) by the human graders. Signs of DR of any degree were found in 69.9% of eyes. One hundred and nine eyes (33.9%) had background DR (R1), and 31.7% (102 eyes) showed pre-proliferative DR (R2). Fourteen eyes (4.3%) had proliferative DR (R3). In 16.5% of eyes, other ocular posterior segment conditions were found, including advanced optic nerve disc cupping (7 eyes, 2.2%), asteroid hyalosis/

Table 1 Patient’s characteristics

<i>n</i> = 165	
Eyes, <i>n</i>	322
Female gender, <i>n</i> (%)	77 (47%)
Age (mean ± SD), years	64.7 ± 14
Diabetes duration, (mean ± SD), years	16.5 ± 12
Type 2 diabetes, <i>n</i> (%)	138 (84%)
Hb1Ac (mean ± SD), %	7.5 ± 1.3
Insulin treatment, <i>n</i> (%)	52 (31%)
Oral treatment, <i>n</i> (%)	154 (93%)
Diabetic diet, <i>n</i> (%)	18 (11%)

n, number; SD, standard deviation

synchysis scintillans (10 eyes, 3.1%), epiretinal membrane/macular pucker (22 eyes, 6.8%) and age-related macular degeneration (19 eyes, 5.9%). Interobserver agreement for RDR was obtained in 97.8% of cases. The most common discrepancy between graders before adjudication was in distinguishing R1 and R2 on the UK NHS severity scale, noted in 11 eyes (3.4%). Once images were uploaded to the EyeArt algorithm, the software automatically detected several specific DR lesions, as shown in Fig. 1.

The distribution of the DR severity is summarized in Table 2. According to the UK NHS classification scale, a patient is deemed non-referable if there are mild or no signs of DR (R0 or R1) and no DME (M0). [18] With human grading as the reference standard, the sensitivity and specificity of AI-based software fed with images captured by each device in the present study is shown in Table 3. In relation to the detection of RDR, similar sensitivity was registered with both devices whereas specificity was higher with the white LED

confocal scanner. The ROC curves for RDR (Fig. 2, Table 3) show that the AUC is higher with the scanner (z statistic = 3.047, $p = 0.002$), implying a better grading accuracy.

Discussion

Early detection and prompt treatment of DR are of great significance to prevent vision loss and promote living quality. Conventional screening methods are dependent on physicians' professional experience and knowledge, they are usually time-consuming and imply a huge waste of medical data. Conversely, deep integration of ophthalmology and AI has the potential to improve the management of patients with DR and easily analyse a massive amount of medical data with the aim to alleviate the social burden and produce a better use of healthcare resources.

Fig. 1 A macula-centred image of a R1M1 diabetic eye acquired with white LED confocal scanner (on the top left) and with conventional flash fundus camera (on the top right). On the bottom, the same images after automated EyeArt detection of haemorrhages and hard exudates. Since both devices produce images with different fields of view, the images were cropped, and automated analysis was considered on the same retinal field size

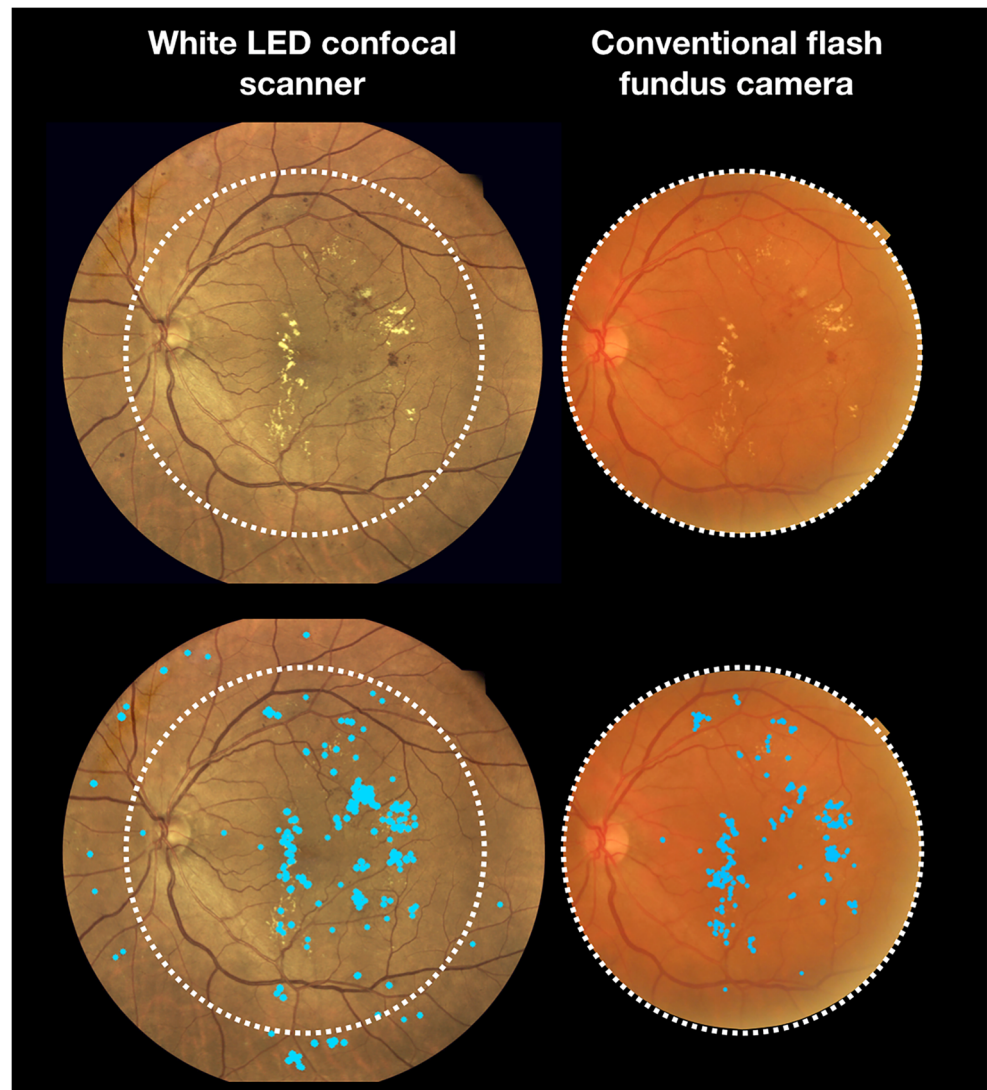


Table 2 Comparison between human grading and AI software using a conventional flash fundus camera or a white LED confocal scanner

	Human grading	EyeArt using conventional flash fundus camera	EyeArt using white LED confocal scanner
R0M0, <i>n</i> (%)	97 (30.1)	83 (25.7)	90 (27.9)
R1M0, <i>n</i> (%)	73 (22.7)	56 (17.4)	63 (19.6)
R1M1, <i>n</i> (%)	36 (11.2)	47 (14.6)	40 (12.4)
R2M0, <i>n</i> (%)	20 (6.2)	30 (9.3)	24 (7.4)
R2M1, <i>n</i> (%)	82 (25.5)	86 (26.7)	85 (26.4)
R3M0, <i>n</i> (%)	9 (2.8)	11 (3.4)	10 (3.1)
R3M1, <i>n</i> (%)	5 (1.5)	5 (1.5)	7 (2.2)
Referable DR, <i>n</i> (%)	152 (47.2)	179 (55.6)	165 (51.2)
No referable DR, <i>n</i> (%)	170 (52.8)	139 (43.2)	154 (47.8)
U, <i>n</i> (%)	n/a	4 (1.2)	3 (0.9)

Analysis performed on 322 eyes

DR, diabetic retinopathy; referable DR (UK NHS scale R1M1; R2M0; R2M1; R3M0; R3M1); no referable (UK NHS scale R0M0; R1M0); *n*, number; U, ungradable; LED, light-emitting diode

The enthusiasm in the field of AI has led to several studies using retinal images to test the performance of automated grading systems for detecting DR. An adequate balance between high sensitivity and specificity is the key to establishing a cost-effective automated screening programme. With low sensitivity, more cases of DR would be missed; that is in contrast with the main scope of a DR screening programme. With unsatisfactory specificity, a relatively large number of false positives would be referred to further examination, resulting in a waste of resources [21].

In 2008, one of the earliest studies on automatic detection of DR from colour fundus photographs reported that an AI-based software was able to detect RDR with 84% sensitivity and 64% specificity [22]. During the years, computing advances and algorithm refinement have led to impressive results that enable AI systems to self-learn and improve. Using databases composed of millions of colour retinal images for training convolutional neural network, various DL algorithms have now surpassed traditional machine-learning methods. This led to substantial improvements in DR detection, achieving significantly higher sensitivity and specificity [10–12].

EyeArt is an AI-based software for detecting DR, and its DL algorithms were trained on hundreds of thousands of cases

and tested in the real world on more than half million patients. Recently, a retrospective study conducted on 107,001 diabetic patients reported that automated DR screening using the EyeArt system achieves high sensitivity (91.3%) and specificity (91.1%) [23]. This algorithm has also been shown to be effective in smartphone camera-based fundus images. In a study of retinal images from 296 patients, which were captured using a portable smartphone-based imaging device, the EyeArt system provided 95.8% sensitivity and 80.2% specificity for detecting any DR and 99.1% sensitivity and 80.4% specificity for detecting sight-threatening DR [24].

Most studies on AI focused mainly on the analysis of conventional flash colour photographs for the automated detection of diabetic eye disease. Different imaging devices, various imaging protocols and intrinsic noise of data can affect the data's quality, which may have influences on models' performance. Therefore, data quality remains a crucial point for grading DR including automatic modalities with AI. Previously, the EyeArt system has proven that sensitivity and false-positive rates are not affected by not only ethnicity and gender but also camera type. Unfortunately, no details are available on the multivariable logistic regression [21].

Table 3 Sensitivity, specificity and AUC of automated grading, as compared with manual grading on the UK NHS scales (*n* = 322)

	EyeArt using conventional flash fundus camera	EyeArt using white LED confocal scanner
Sensitivity, % (95% CI)	90.8 (95% CI 85.0–94.9)	94.1 (95%CI: 89.1–97.3)
Specificity, % (95% CI)	75.3 (95% CI 68.0–81.7)	86.8 (95%CI: 80.7–91.6)
AUC, % (95% CI)	0.830 (95% CI 0.78–0.87)	0.905 (95%CI: 0.87–0.93)
Difference between AUC (95%CI, <i>p</i> value)	0.0737 (95% CI 0.0263–0.121), <i>p</i> = 0.0023	

AUC, area under the receiver operating characteristic curve; 95% CI, 95% confidence interval; LED, light-emitting diode; UK NHS, UK National Health Service screening programme

Advances in retinal imaging techniques have revolutionized the fundus examination in ophthalmology. For example, the Optos fundus camera (Optos PLC, Dunfermline, UK) is a confocal scanning laser ophthalmoscope that provides 200-degree ultra-widefield pseudocolour imaging. Recently, Wang and colleagues conducted a study to evaluate the effectiveness of EyeArt software for detecting RDR in Asian Indian patients using Optos pseudocolour images. In this study, the authors observed a good sensitivity (91.7%), but a moderate specificity (50%) for the detection of referral-warranted DR. According to the authors, this low specificity value may be related to the fact that the algorithm was not optimized for UWF images and the pseudo coloration provided by Optos images may have fooled the algorithm in detecting non-RDR [25].

The Eidon scanner combines confocal technology with white LED illumination in a fully automated mode. In comparison with conventional flash fundus camera, the scanner offers higher-resolution and higher-quality true-colour retinal images [17].

Available evidence shows that accuracy by AI systems may not be influenced by different retina imaging systems based on the same technology (flash fundus cameras).

Conversely, different imaging technologies (LED confocal scanner, UWF scanning laser ophthalmoscopy, smartphone cameras) may result in different accuracy when using an automated AI analysis system.

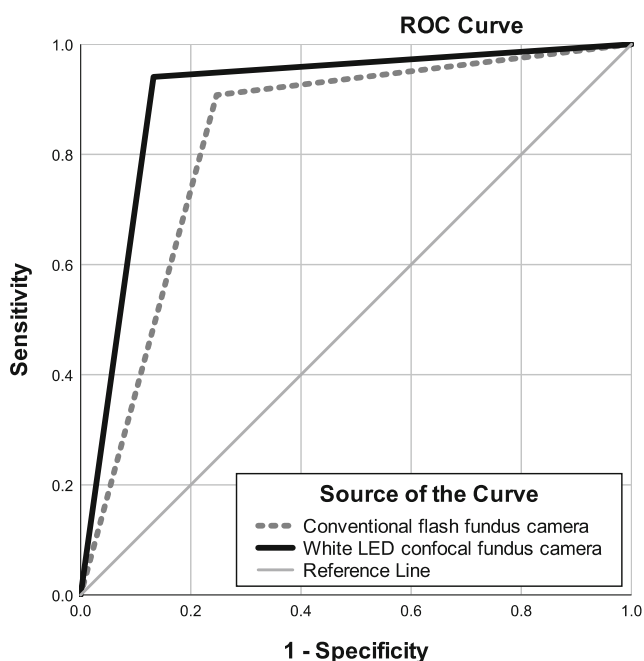


Fig. 2 ROC curves for referable diabetic retinopathy. The broken line represents the ROC curve of EyeArt algorithm using white LED confocal scanner. The continuous black line represents the ROC curve of EyeArt algorithm using the conventional flash fundus camera. Continuous grey line is shown as reference line

However, the clinical significance of this difference in the context of screening campaigns should be further evaluated, especially when the variability is related to specificity such as in this study. The present study shows that, compared with human grading, EyeArt software achieved high sensitivity for RDR using both conventional flash fundus camera (90.8%) and white LED confocal scanner (94.1%). A higher specificity was recorded using the scanner (86.8%) resulting a superior diagnostic accuracy in this setting.

This difference may be related to three factors. First, the two devices provide images with different resolution. Unlike the flash fundus camera, the scanner captures 14 megapixel retinal images. It was previously observed that the AUC performance of human graders for detecting DR increases with the input image size, which could be attributed to a larger amount of information and features [26]. A second aspect is related to the confocal attribute. By using a slit confocal system, light reflected from out-of-focus layers is masked and provides only marginal contribution to image formation. This reduces the choroidal contribution to the final picture, enhancing the contrast of the resulting image, limiting the noise and finally improving its discriminative power [17]. A third factor is related to the chromaticity properties of the scanner. In a previous study, we demonstrated that it provides balanced colour images with a wider richness of colour content compared with conventional flash fundus camera [17].

Undoubtedly, our study has some limitations, which should be considered when assessing the results. First, the sample size is relatively small, and the proportion of patients with R2 is higher than what typically reported in a screening programme. Second, the images provided by the white LED confocal scanner were cropped such that they closely fit the 45-degree field of view in order to perform a fair comparison between the two imaging modalities. This may have limited the overall performance of the AI-based algorithm when using images obtained from the scanner. Third, the EyeArt software was largely trained with conventional flash fundus cameras, and this might have an influence on the results obtained with the scanner. Fourth, in this study, the diagnostic performance of the AI algorithm was calculated based on the results for each eye (eye level). Results at the eye level are more relevant to assess raw algorithm performance, as per the purpose of the present study. However, the combined results for both eyes (patient level) are more relevant to the clinical scenario and therefore are to be preferred when evaluating the performance of large-scale screening programme in future studies. Moreover, only one macula-centred image was acquired for each eye by both devices. There is high level of evidence that single-field fundus photographs can serve as a screening tool for DR to identify patients with RDR providing a sensitivity ranging from 61 to 90% and specificity ranging from 85 to 97% when compared with the 7 standard fields [27]. However, it must be taken into account that lots of information

about the retinal periphery could be missed when the automated DR analysis is focused exclusively on the posterior pole. Fifth, this study included only Caucasian patients. Therefore, it is uncertain if our results can be generalized to other populations. The implementation of demographic data, including gender, ethnicity, type and duration of diabetes, levels of haemoglobin A1c (HbA1c) and blood pressure, could improve the outcomes of an automated DR screening campaign. Moreover, as the AI system did not use data on visual acuity, the specificity observed for the AI system is reduced.

Another aspect that should be considered is that this study, as most experiences on automatic diagnosis of eye diseases, focused on binary classification, whereas in a clinical setting, patients typically suffer from multicategorical retinal conditions. According to a study conducted by Choi et al., the accuracy of a DL-based algorithm tends to decline from 87.4 down to 30.5% as the number of retinal diseases considered increases [28]. In our study population, 16.5% of patients had other concomitant retinal conditions, and this may have influenced our results.

In conclusion, despite the cited limitations, our study shows that EyeArt is well suited to work with different imaging technologies, such as conventional flash fundus camera or white LED confocal scanner. Further studies on large scale are needed to evaluate how technologies alternative to conventional fundus cameras, with higher image resolution and colour content, may impact the performance of AI-automated analysis.

Authors' contributions V. Sarao and D. Veritti contributed equally to this manuscript and share the first authorship on this work. All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by V. Sarao and D. Veritti. The first draft of the manuscript was written by V. Sarao and D. Veritti and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability The data used to support the findings of this study are available from the corresponding author upon request.

Compliance of ethical standard

Conflict of interest V. Sarao has acted as a consultant for Centervue; D. Veritti has acted as a consultant for Novartis and Roche; P. Lanzetta has acted as a consultant for Allergan, Bayer, Centervue, Novartis and Roche.

Ethics approval General Data Protection Regulation (GDPR) compliance was maintained. Care of the patients in this study was in accordance with the Declaration of Helsinki.

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